

Manifold Topology and In-Context Out-of-Prior Detection

Anonymous submission

Manifold Topology and In-Context Out-of-Prior Detection

Prior-Data Fitted Networks (PFNs) implement approximate Bayesian inference in their forward pass. When the input context dataset $D_c = \{(x_i, y_i)\}_{i=1}^T$ deviates from the training prior $\mathcal{P}_{ID}$, the network’s behavior is mathematically undefined. To detect such out-of-prior (OOD) contexts, we examine the internal hidden representations of the model.

Representation Extraction and Subspace Projection

Let $z \in \mathbb{R}^d$ ($d = 256$) be the sequence-averaged representation extracted from the final Transformer encoder layer (Layer 5) after processing the context dataset D_c . To isolate the structural prior boundary from high-dimensional observational noise, we project z into a low-dimensional subspace using a non-linear projector $\Phi : \mathbb{R}^d \rightarrow \mathbb{R}^2$ (implemented via UMAP).

Crucially, the projector Φ is fit jointly on a union of N_{cal} in-distribution calibration samples and N_{ref} out-of-prior reference samples:

$$X_{fit} = X_{cal} \cup X_{ood.ref} \quad (1)$$

This joint fitting prevents the collapse of OOD-discriminating dimensions that occurs when fitting only on in-distribution data.

Mahalanobis Distance Metric

In the projected space, the out-of-prior scoring function $S(D_c)$ is defined as the Mahalanobis distance relative to the projected calibration set:

$$S(D_c) = \sqrt{(\Phi(z) - \mu_{cal})^T \Sigma_{cal}^{-1} (\Phi(z) - \mu_{cal})} \quad (2)$$

where μ_{cal} and Σ_{cal} are the mean vector and covariance matrix of the projected calibration set $\Phi(X_{cal})$, respectively.

The Five-Suite Evaluation Framework

To systematically evaluate OOD detection, we construct five test suites ($N = 1000$ context samples each):

1. **ID, Low Noise:** Spectral Mixture (SM) GP samples generated from the training prior with low noise ($\sigma \leq 0.001$). This serves as the clean negative test set.
2. **ID, High Noise:** SM GP samples from the training prior corrupted by high observation noise ($\sigma \sim 0.4$). This checks whether the detector erroneously fires on noise.
3. **Near-OOD:** SM GP samples with frequency parameters shifted outside the training range ($f_{max} \sim [5.0, 10.0]$).
4. **Far-OOD:** SM GP samples with frequency parameters shifted extremely far outside the training range ($f_{max} \sim 15.0$).
5. **Real-World OOD:** Normalized segments of real-world regression time series (e.g., airline passenger counts and minimum daily temperatures).

Experimental Results and the Noise Robustness Paradox

We evaluate UMAP-2D Mahalanobis distance against two key baselines: Mahalanobis distance in a linear PCA-10D subspace, and the in-context prediction mean squared error (MSE) on query points. The quantitative metrics are shown in Table 1, and the corresponding ROC curves are plotted in Figure 2.

Our results reveal a critical trade-off that we term the *Noise Robustness Paradox*. In-context prediction error (MSE) and linear representations (PCA-10D) are highly sensitive to out-of-prior data. However, they achieve AUROC scores of 1.000 and 0.953 respectively on the *Robustness to Noise* task. This means they cannot distinguish structural prior violation from simple observation noise, triggering false alarms on noisy ID datasets.

Conversely, the joint-fitted UMAP-2D Mahalanobis distance remains close to chance-level performance on noisy ID data (AUROC = 0.471), demonstrating complete robustness to observation noise. At the same time, it maintains strong detection performance on Near-OOD (AUROC = 1.000), Far-OOD (AUROC = 1.000), and Real-World OOD (AUROC = 0.840). The non-linear projection successfully filters out high-frequency noise variance while preserving the topological features representing prior violation.

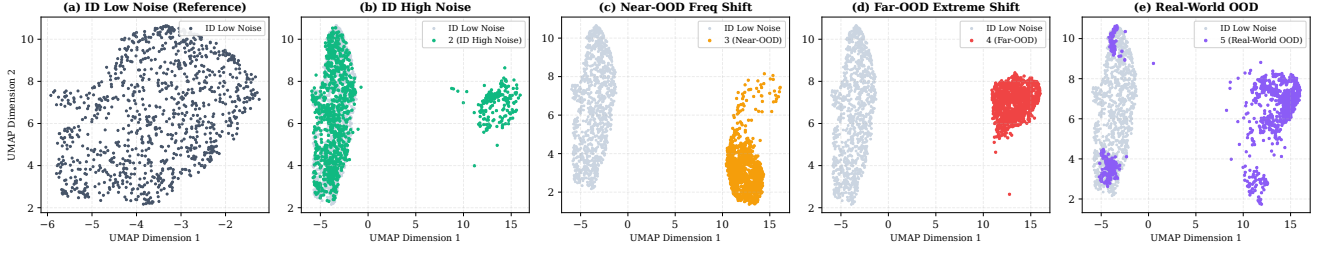


Figure 1: Joint-fitted UMAP projection of Layer 5 activations showing (a) the pure calibration manifold reference set (blue), and (b–e) comparisons against the four evaluation test suites. High observation noise (b) remains overlapping with the calibration manifold, while Near-OOD frequency shift (c), Far-OOD extreme frequency shift (d), and Real-World OOD (e) show distinct topological separation.

Table 1: Quantitative evaluation of PFN Out-of-Prior detection. UMAP-2D Mahalanobis balances high detection power with noise robustness (chance-level AUROC of 0.471 on ID High Noise).

Evaluation Task	UMAP-2D Mahalanobis (Ours)			PCA-10D Mahalanobis			Prediction MSE (Baseline)		
	AUROC $\uparrow$	AUPR $\uparrow$	FPR@95 $\downarrow$	AUROC $\uparrow$	AUPR $\uparrow$	FPR@95 $\downarrow$	AUROC $\uparrow$	AUPR $\uparrow$	FPR@95 $\downarrow$
Robustness to Noise (ID High Noise)	0.526	0.618	0.965	0.983	0.985	0.080	1.000	1.000	0.000
Near-OOD Detection (Freq Shift)	1.000	1.000	0.000	1.000	1.000	0.000	1.000	1.000	0.000
Far-OOD Detection (Extreme Freq)	1.000	1.000	0.000	1.000	1.000	0.000	1.000	1.000	0.000
Real-World OOD Detection	0.882	0.920	0.687	1.000	1.000	0.000	1.000	1.000	0.000

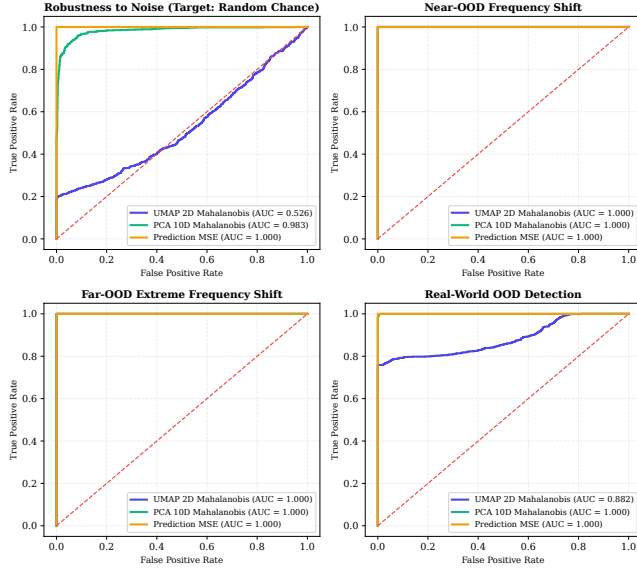


Figure 2: OOD detection ROC curves. While prediction MSE and PCA-10D achieve high OOD sensitivity, they fail the noise robustness check by incorrectly classifying high-noise ID samples as OOD ($\text{AUROC} \geq 0.95$). UMAP-2D Mahalanobis remains blind to noise ($\text{AUROC} \approx 0.47$) while maintaining high detection power.